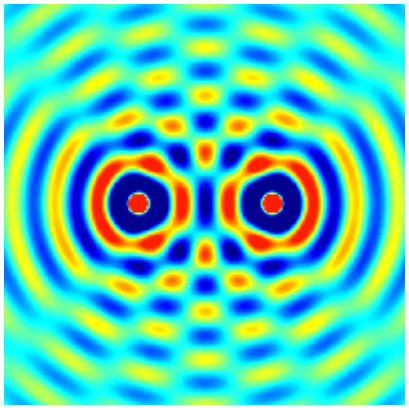
§4 两个点源的干涉

§4.1 两列球面波的干涉场

§4.2 杨氏双缝干涉实验

§4.3 两束平行光的干涉场



§4.1 两列球面波的干涉场

问题： 设在空间中有两个点源Q_1和Q_2，初始相位分别为φ_{10}和φ_{20}，频率相同，振动方向平行，相距为d，求空间中任一场点P的光强

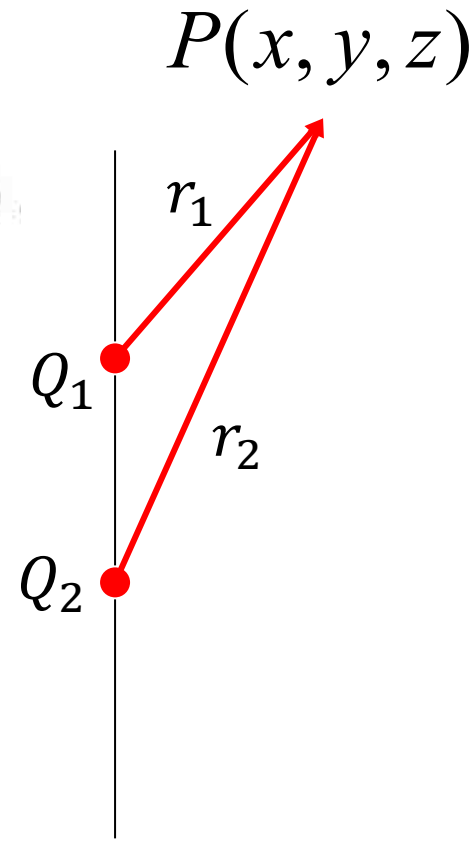
P的光强可表示为

$$I(P)=I_1(P)+I_2(P)+2\sqrt{I_1(P)I_2(P)}\cos\delta(P)$$

其中 $I_1(P)=[A_1(P)]^2$

$I_2(P)=[A_2(P)]^2$

振幅与距离成反比



当 $d \ll r_1$，r_2时，则

$$A = A_1(P) \approx A_2(P)$$

则光强可进一步表示为

$$I(P)=2A^2[1+\cos\delta(P)]$$
$$=4A^2\cos^2\frac{\delta(P)}{2}.$$

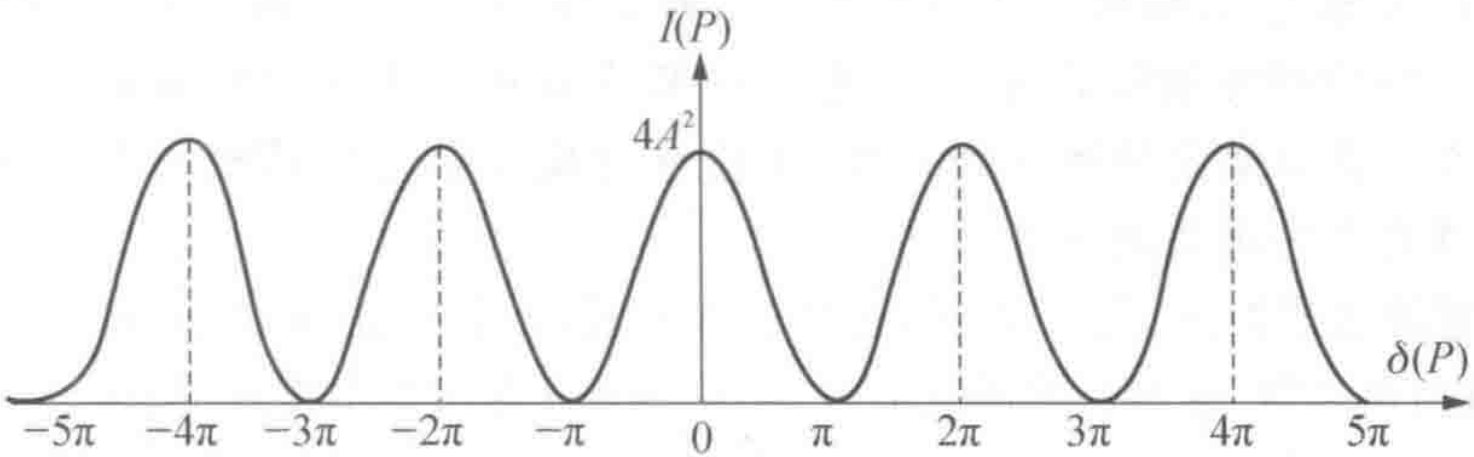


图 4-2　两波相干叠加时强度与相位差的关系

我们来考察相位差$\delta(P)$

$$\delta(P)=\varphi_1(P)-\varphi_2(P)$$

其中

$$\begin{cases}\varphi_1(P)=\varphi_{10}+kr_1=\varphi_{10}+\dfrac{2\pi}{\lambda}r_1,\\ \varphi_2(P)=\varphi_{20}+kr_2=\varphi_{20}+\dfrac{2\pi}{\lambda}r_2.\end{cases}$$

注意非真空的情况

从而得到 $$\delta(P)=\varphi_{10}-\varphi_{20}+\frac{2\pi}{\lambda}(r_1-r_2)$$

为方便起见，设$\varphi_{10}-\varphi_{20}=0$ $\longrightarrow$ $$\delta(P)=\frac{2\pi}{\lambda}(r_1-r_2)$$

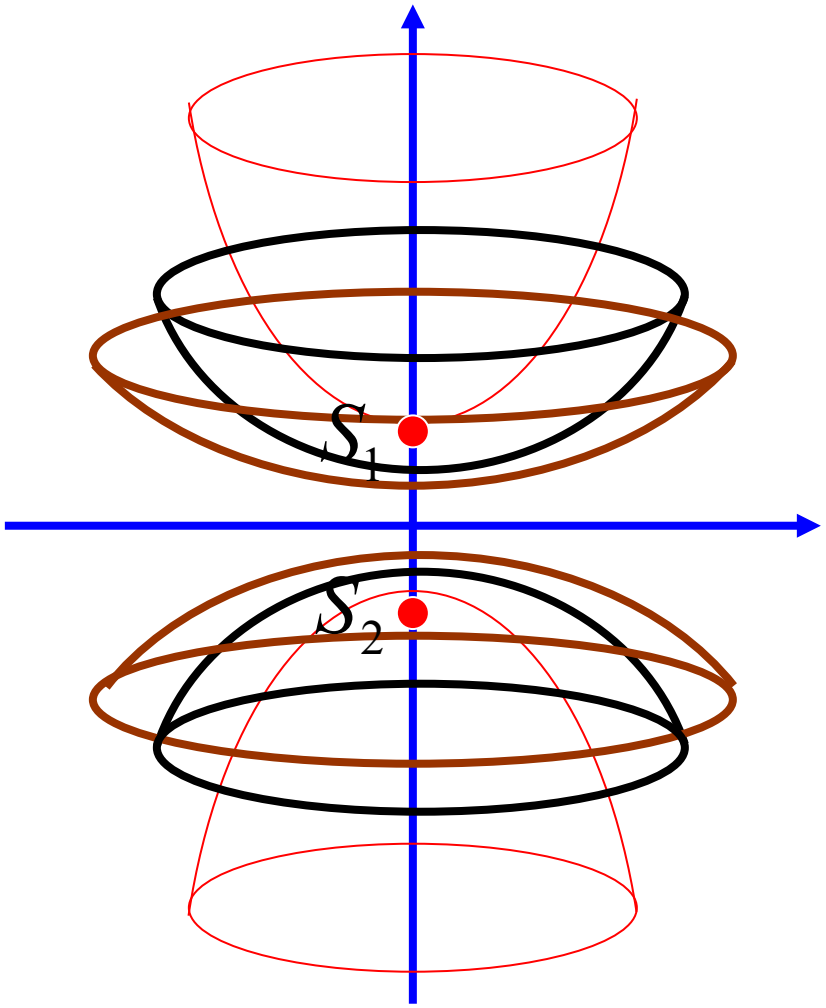
令光程差$\Delta L = r_1 - r_2$

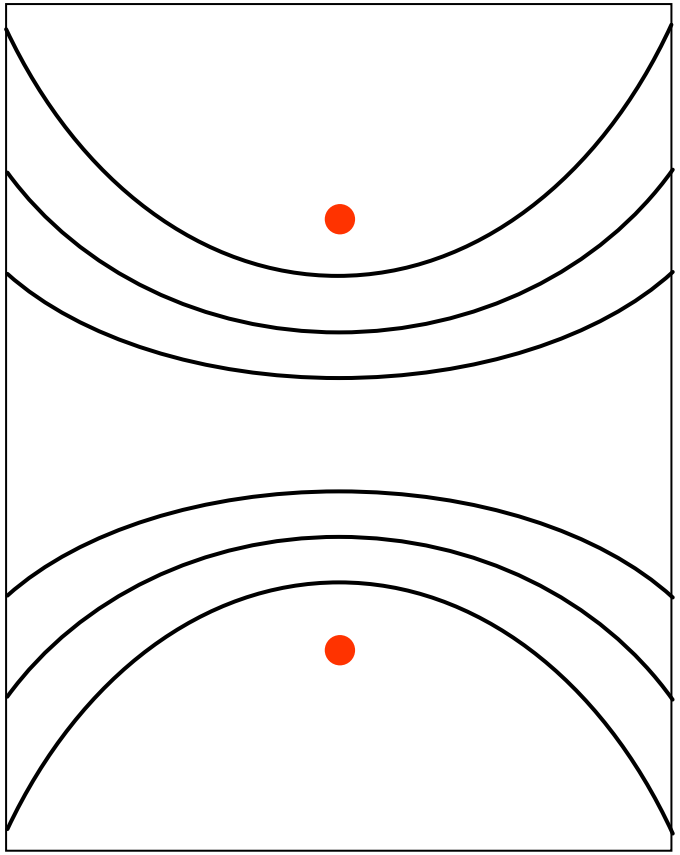
$$I(P) = 4A^2 cos^2 \frac{\delta(P)}{2}$$ 和 $$\delta(P) = \frac{2\pi}{\lambda}(r_1 - r_2)$$

得到光强极大、极小的条件为

$$\begin{cases} \text{极大} \quad \Delta L = k\lambda, \\ \text{极小} \quad \Delta L = \left(k + \frac{1}{2}\right)\lambda \end{cases} \quad (k = 0, \pm 1, \pm 2, \cdots)$$

等强面为回转双曲面，接收屏上为双曲线，明暗交错分布，干涉条纹是非定域的，空间处处可见。





等强面为回转双曲面，接收屏上为双曲线，明暗交错分布，干涉条纹是非定域的，空间处处可见。

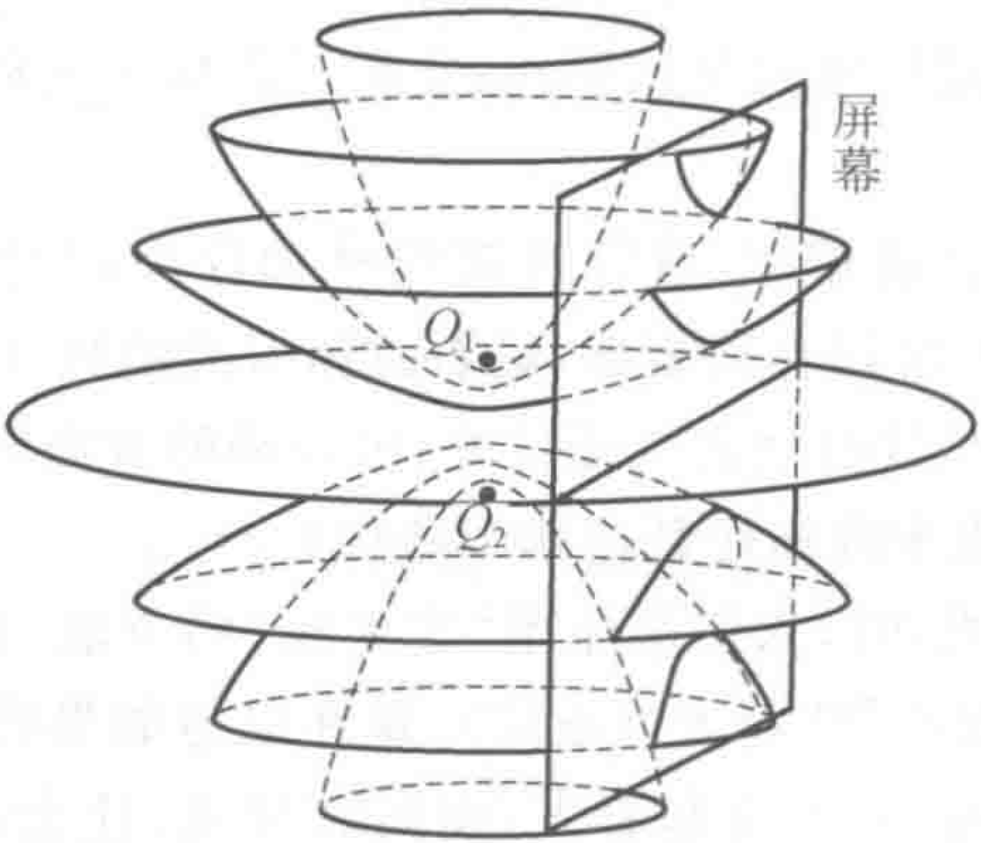
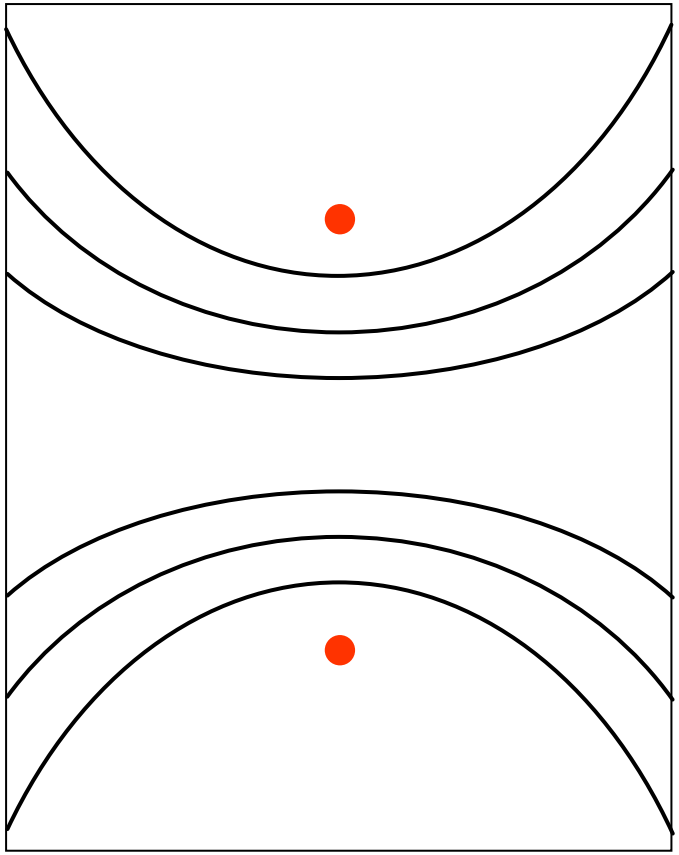
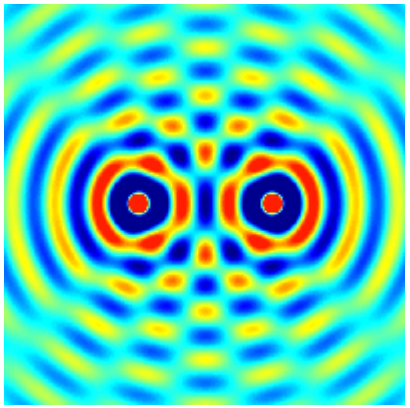


图 4-1　两球面波的干涉场





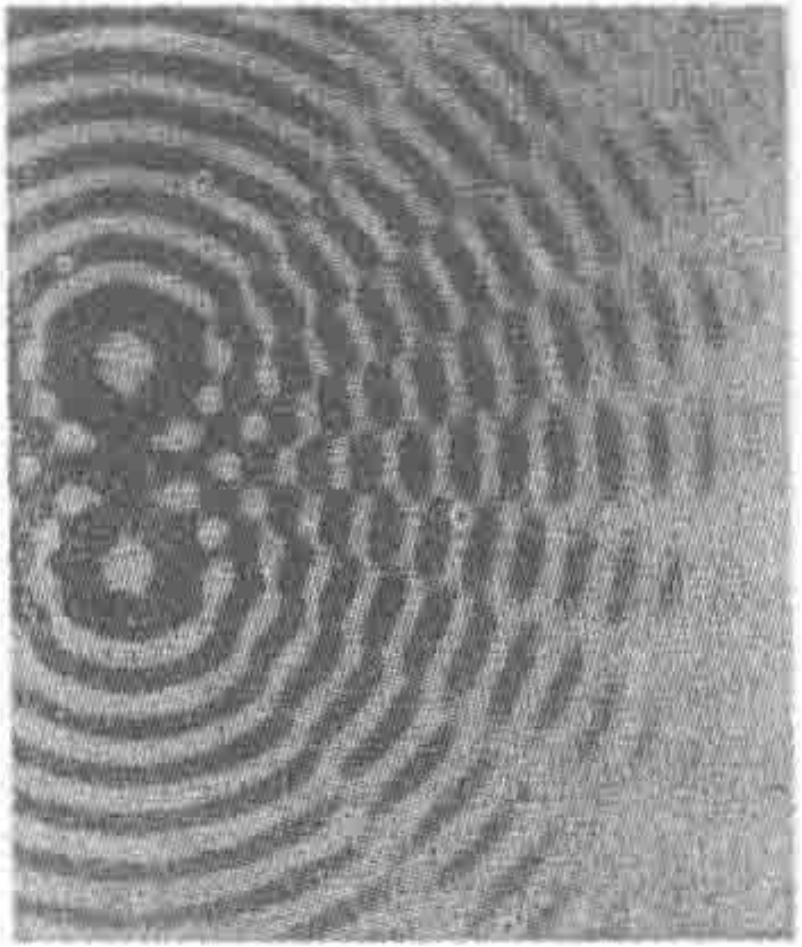


图 4-3　两点源干涉场的水波盘演示

相干条件

归纳起来，产生干涉的必要条件（相干条件）有三条：

(1)频率相同；

(2)存在相互平行的振动分量；

(3)相位差 $\delta(P)$稳定.

其中相位差

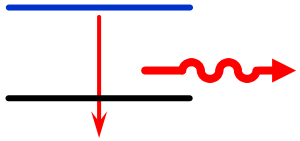
$$\delta(P,t) = (\omega_1 - \omega_2)t - (\varphi_1(P) - \varphi_2(P))$$

3.3 普通光源发光微观机制的特点

发光机制：自发辐射，受激辐射

普通光源：自发辐射，间歇性，发光时间~10^{-8}秒

随机性



几个重要的时间间隔：

i） 光扰动的时间周期：$T \sim 10^{-15}$ sec

ii）实验观测的时间（人眼的响应时间）：$\tau \sim 10^{-1}$ sec

iii）探测器响应时间：$\Delta t \sim 10^{-9}$ sec

由于 $\tau >> \Delta t >> T$ ，所以无论是实际观察还是仪器接收，得到的都不可能是某一瞬间的扰动分布，而只能是扰动强度的时间平均值，即强度。

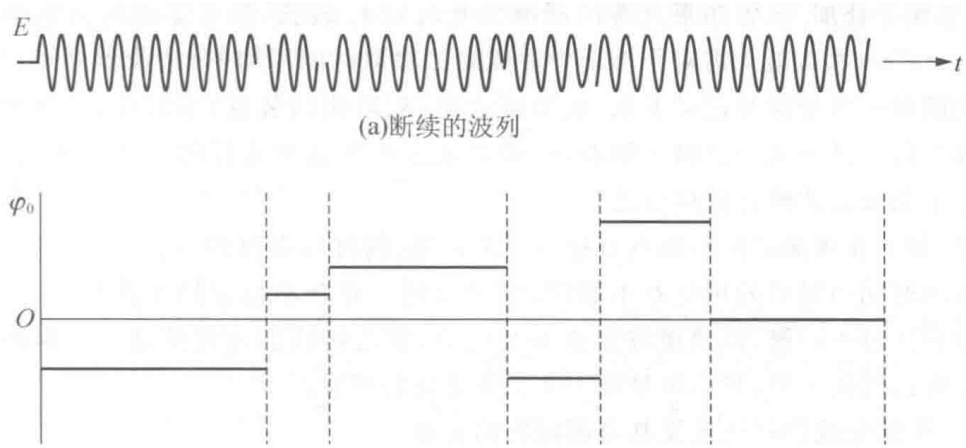
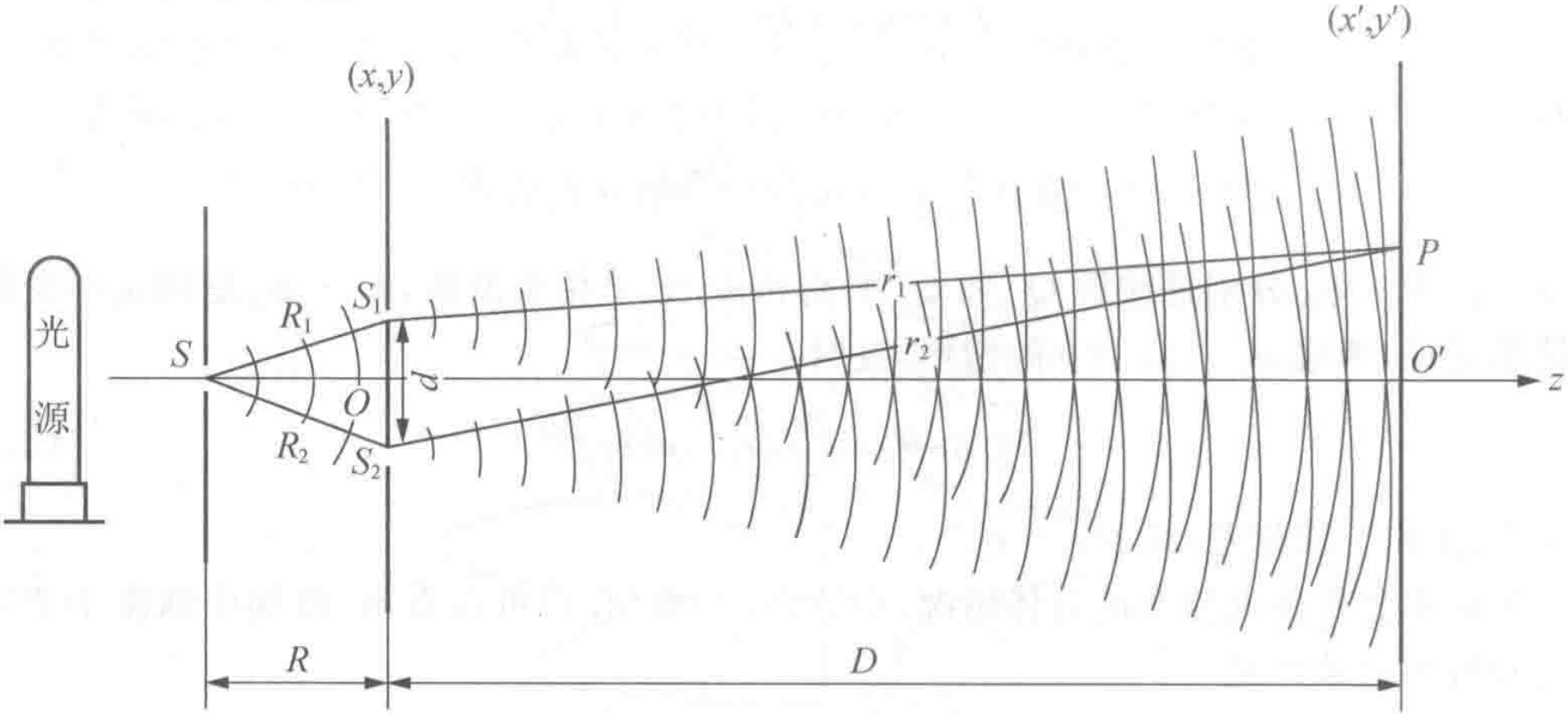


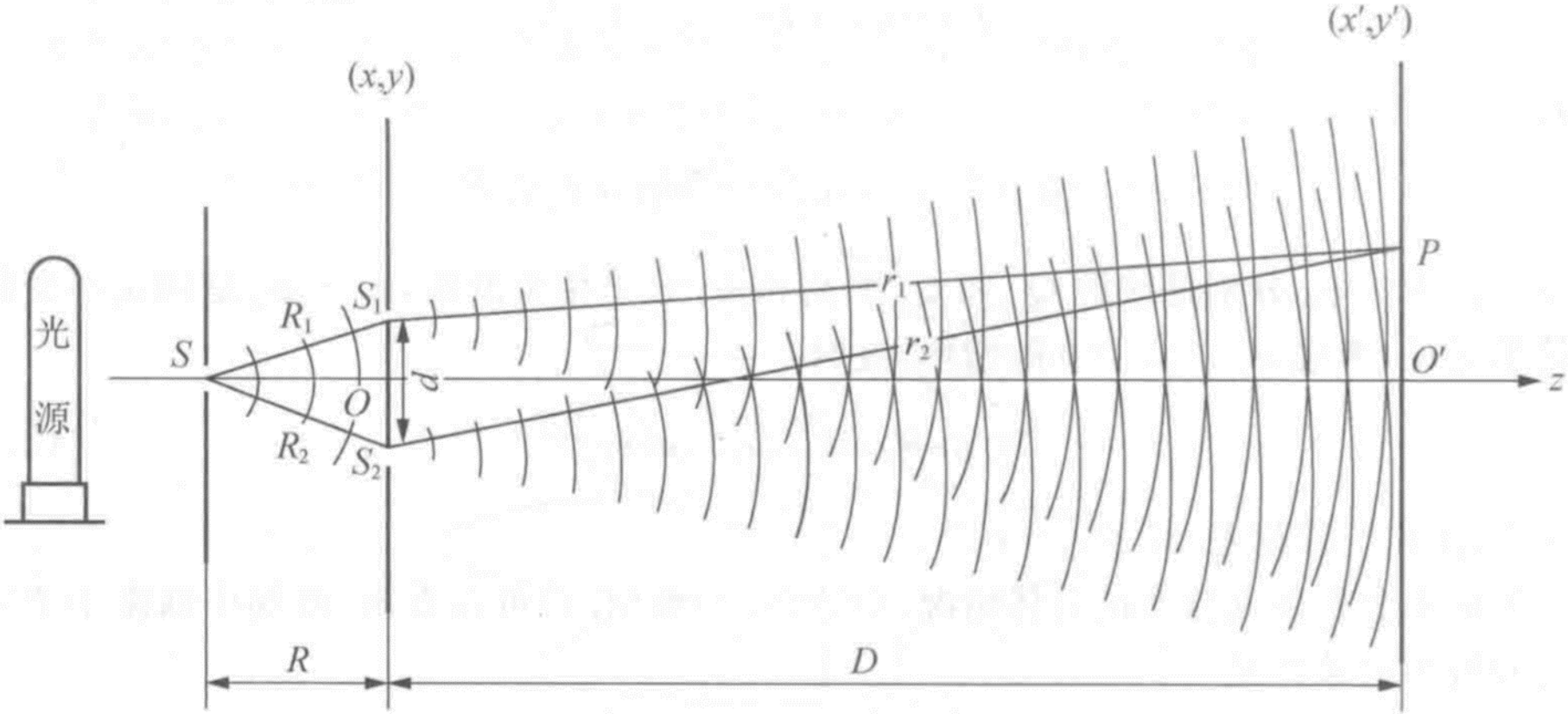
图 3-4 非激光光源中原子发射光波的图像

§4.2 杨氏干涉实验



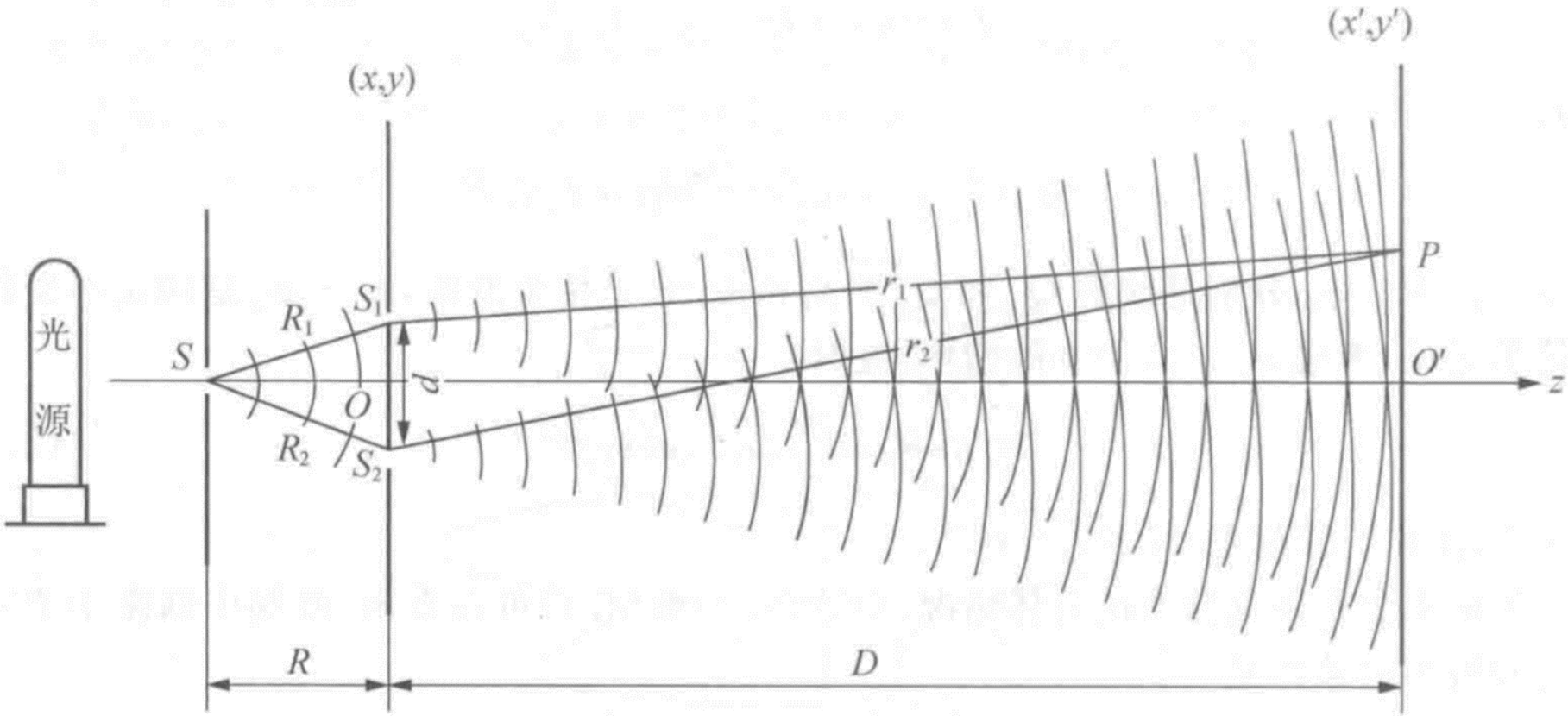
设φ_0是点源S的初相位，则点源S_1和S_2的初相位分别为

$$\varphi_{10}=\varphi_0(t)+\frac{2\pi}{\lambda}R_1,\quad \varphi_{20}=\varphi_0(t)+\frac{2\pi}{\lambda}R_2,$$



则有恒定的初始相位差

$$\varphi_{10}-\varphi_{20}=\frac{2\pi}{\lambda}(R_1-R_2)=0$$



$S_1\left(\frac{d}{2},0\right)$

$S_2\left(-\frac{d}{2},0\right)$

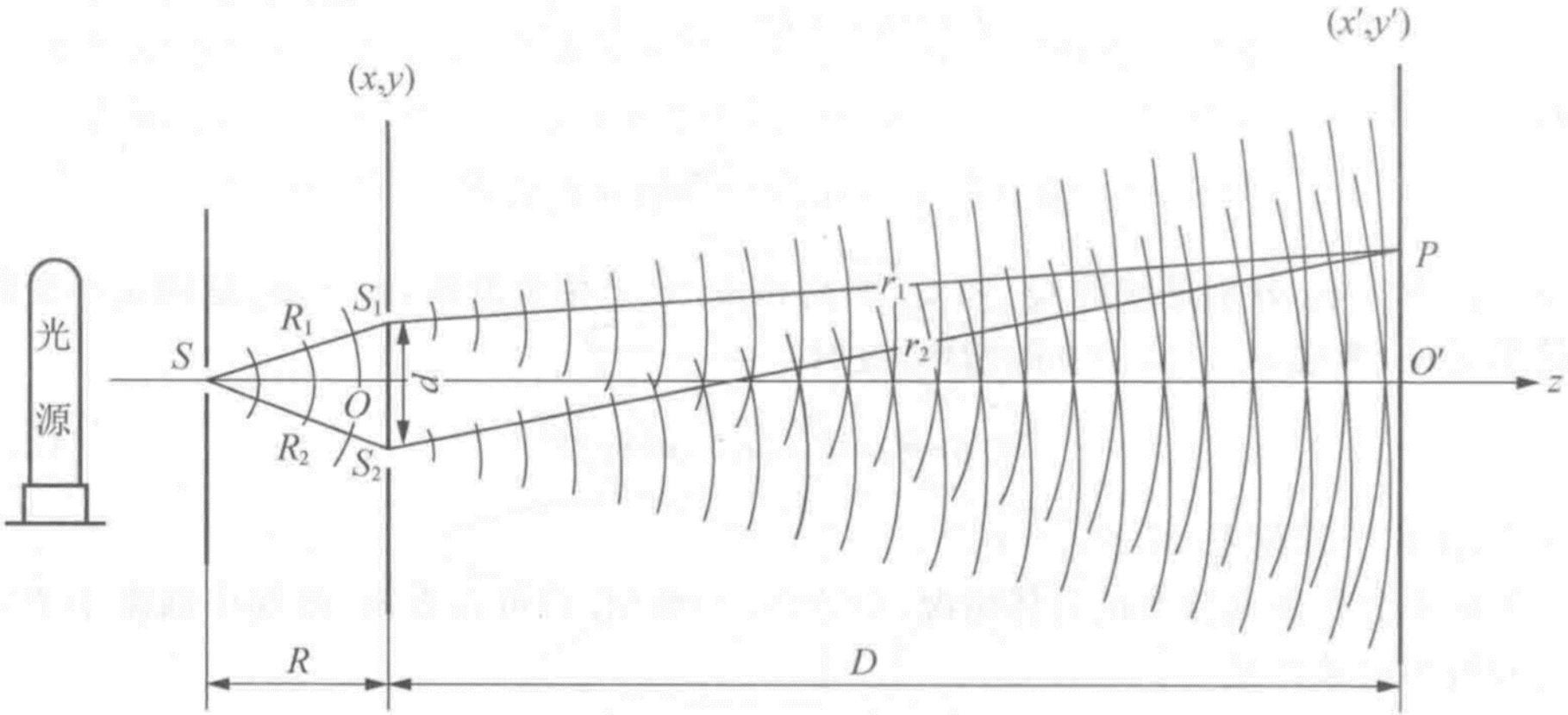
$P(x',y')$

双孔间隔 $d\sim 0.1\text{mm}—1\text{mm}$,

横向观测范围 $\rho\sim 1\text{cm}—10\text{cm}$,

幕与双孔屏的距离 $D\sim 1\text{m}—10\text{m}$.

$$z=D,x=\pm d/2,y=0,r_0'=D+\frac{(d/2)^2}{2D}$$



$$\begin{cases}\tilde{U}_1(x',y')=\dfrac{a}{D}\exp\left[\mathrm{i}k\left(D+\dfrac{(d/2)^2+x'^2+y'^2}{2D}\right)\right]\exp\left(\dfrac{-\mathrm{i}kd}{2D}x'\right),\\ \tilde{U}_2(x',y')=\dfrac{a}{D}\exp\left[\mathrm{i}k\left(D+\dfrac{(d/2)^2+x'^2+y'^2}{2D}\right)\right]\exp\left(\dfrac{\mathrm{i}kd}{2D}x'\right),\end{cases}$$

则合成振幅为

$$\tilde{U}(x',y')=\tilde{U}_1(x',y')+\tilde{U}_2(x',y')$$
$$=\frac{2a}{D}\exp\left[ik\left(D+\frac{(d/2)^2+x'^2+y'^2}{2D}\right)\right]\cos\left(\frac{kd}{2D}x'\right)$$

则强度分布为

$$I(x',y')=\tilde{U}^*(x',y')\tilde{U}(x',y')=4A^2\cos^2\left(\frac{kd}{2D}x'\right)$$

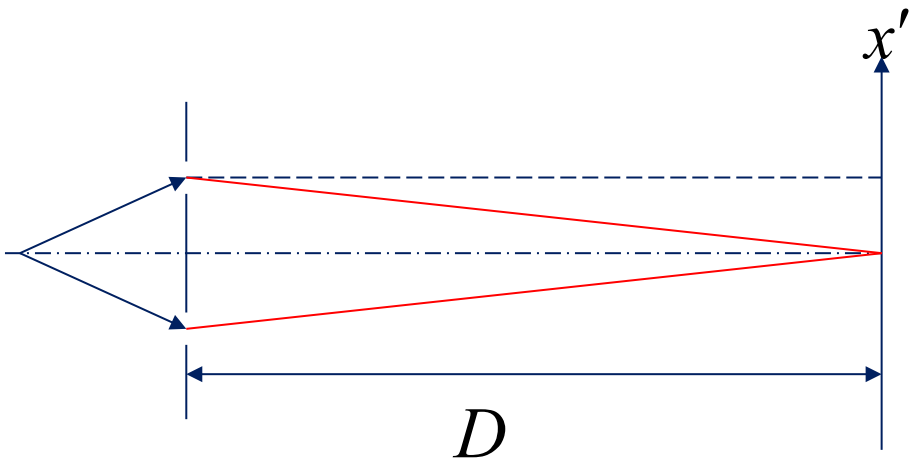
其中

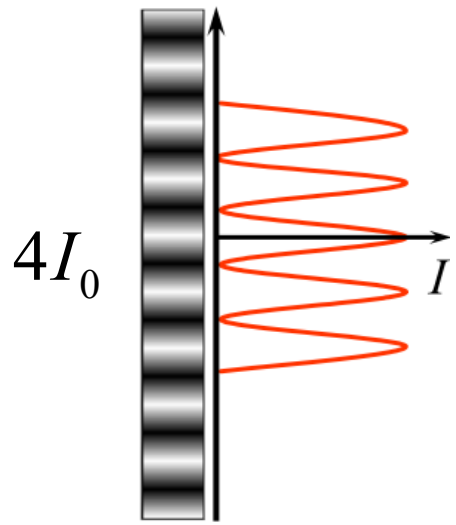
$$A=a/D$$

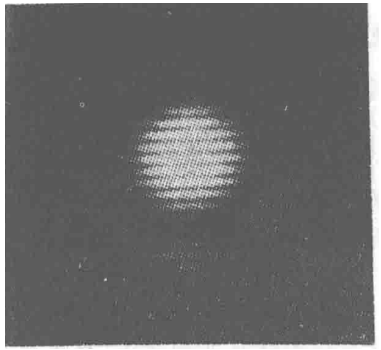
强度分布：

$$I(x',y')=\left(\frac{2A}{D}\right)^2\cos^2(\frac{kd}{2D}x')=4\left(\frac{A}{D}\right)^2\cos^2(\frac{kd}{2D}x')=4I_0\cos^2(\frac{kd}{2D}x')$$

其中： $I_0=(\frac{A}{D})^2$ 从一个孔中出射的光波在屏中心的强度



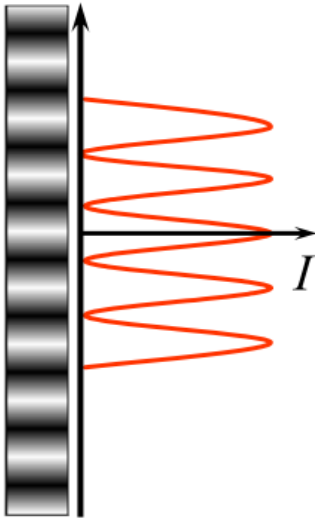




利用次光源分解，巧妙地获得了相干光源!

条纹的特征

强度的空间分布 $I(x',y') = 4I_0\cos^2(\frac{kd}{2D}x')$



i）干涉条纹的形状：

在傍轴条件下，等强线是一组与y’轴平行的直线

干涉相长
(亮条纹)

$$\frac{kd}{2D}x' = j\pi \Rightarrow x' = j\pi\frac{2D}{kd} = j\frac{D}{d}\lambda$$

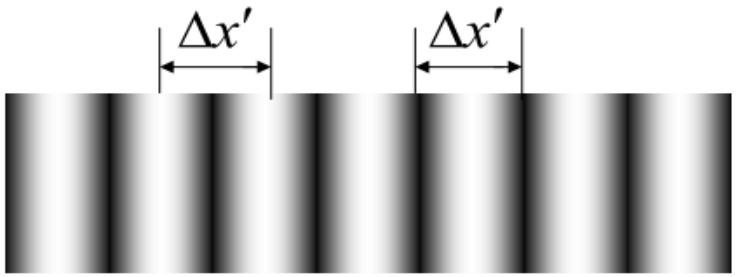
干涉相消
(暗条纹)

$$\frac{kd}{2D}x' = (2j+1)\frac{\pi}{2} \Rightarrow x' = (2j+1)\frac{\pi}{2}\frac{2D}{kd} = \frac{2j+1}{2}\frac{D}{d}\lambda$$

条纹的特征

强度的空间分布 $I(x',y') = 4I_0\cos^2(\frac{kd}{2D}x')$

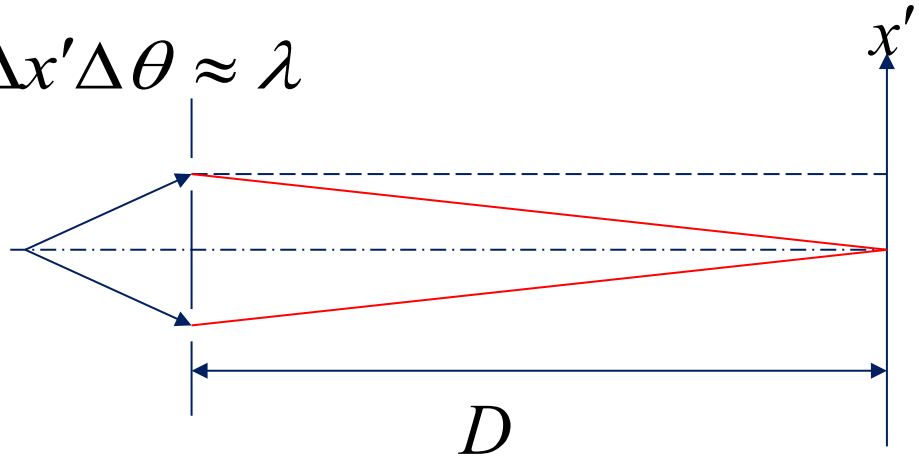
ii）干涉条纹间距



$\Delta x' = \lambda D/d$， $\Delta\theta \approx d/D$ 双缝对场点的张角

$\Delta x' \approx \lambda/\Delta\theta$， 或 $\Delta x'\Delta\theta \approx \lambda$

反比例关系



例1：在杨氏双缝实验中，HeNe激光（λ=632.8nm）照射间隔0.5mm的双孔，2m处平面上干涉条纹的间距，是波长的多少倍

解 $\lambda = 632.8nm \approx 0.6\mu m, \quad d = 0.5mm, \quad D = 2m$

2m处平面上干涉条纹的间距

$$\Delta x' = \lambda D/d \approx 2.4mm$$

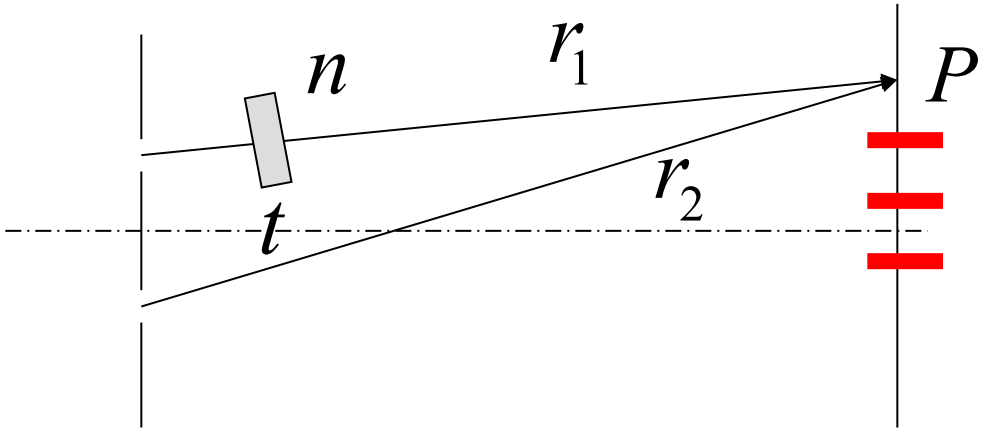
$$\frac{\Delta x'}{\lambda} = D/d = 4\times 10^3 \text{倍}$$

例2：（1）光源和接收屏之间充满介质

$$x' = j\frac{D}{d}\frac{\lambda}{n}$$

相邻亮（暗）条纹间隔 $\Delta x' = \frac{D}{d}\frac{\lambda}{n}$

(2) 在其中一个狭缝处放入云母片



$$r_1' = r_1 - t + nt$$
$$= r_1 + (n-1)t$$
$$\Delta L(P) = (r_1 - r_2) + (n-1)t$$

光程差每改变1个波长，条纹移动1个间隔

思考：加入云母片之后，条纹发生什么变化?

例3：存在两种不同频率的单色光

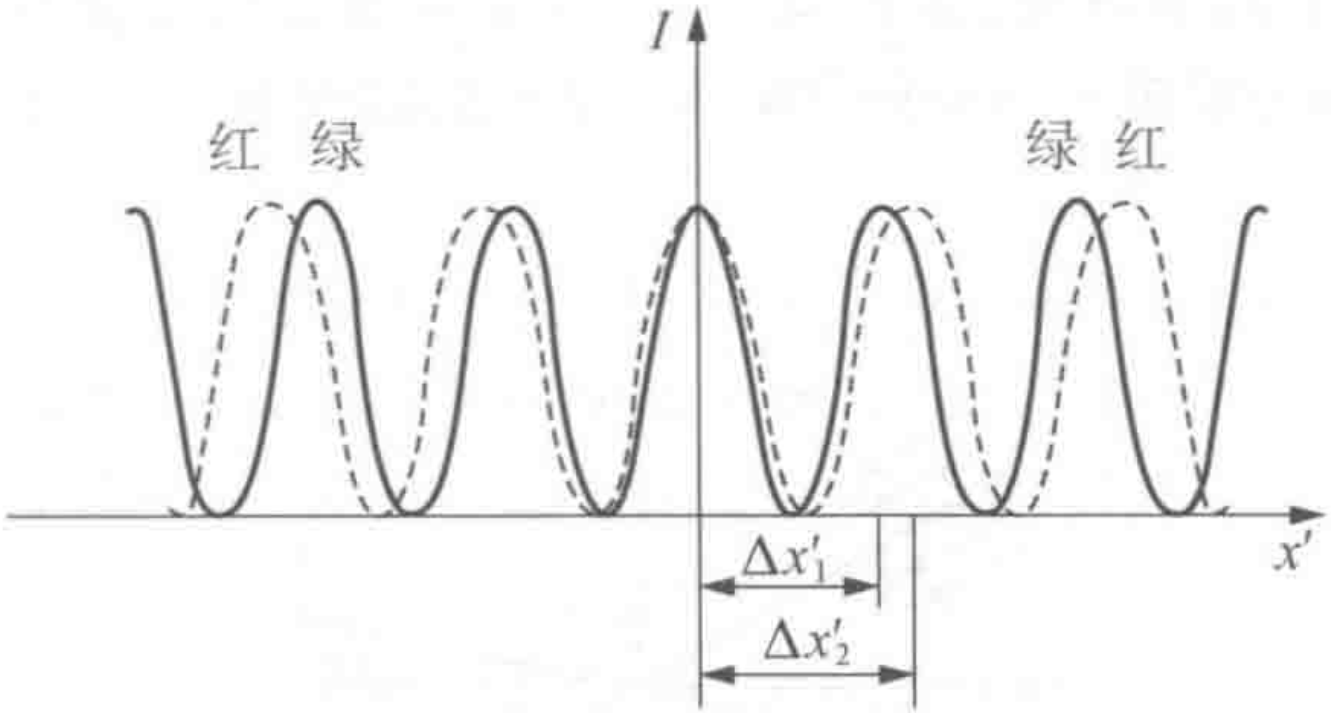


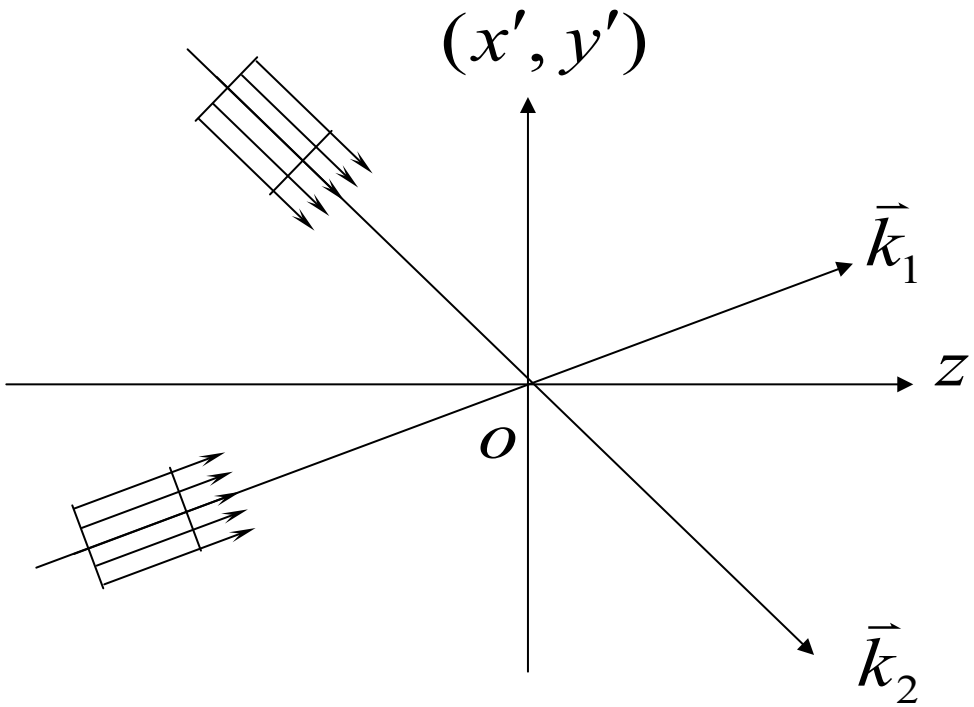
图 4-5　两套不同颜色的干涉条纹不相干叠加

§4.3 两束平行光的干涉场

问题：两列同频率单色光照射在z=0的波前上，振幅分别为A_1，A_2；在0点的初相位分别为φ_{10}，φ_{20}，

两束平行光束的传播方向角为：

$$(\alpha_1, \beta_1, \gamma_1), \quad (\alpha_2, \beta_2, \gamma_2),$$



$$\boldsymbol{k} = k(\cos\alpha \boldsymbol{e}_x + \cos\beta \boldsymbol{e}_y + \cos\gamma \boldsymbol{e}_z) \qquad \boldsymbol{r} = x\boldsymbol{e}_x + y\boldsymbol{e}_y + z\boldsymbol{e}_z$$

在 z=0 的波前上:

$$\begin{cases} \varphi_1(x', y') = k(x'\cos\alpha_1 + y'\cos\beta_1) - \varphi_{10} \\ \varphi_2(x', y') = k(x'\cos\alpha_2 + y'\cos\beta_2) - \varphi_{20} \end{cases}$$

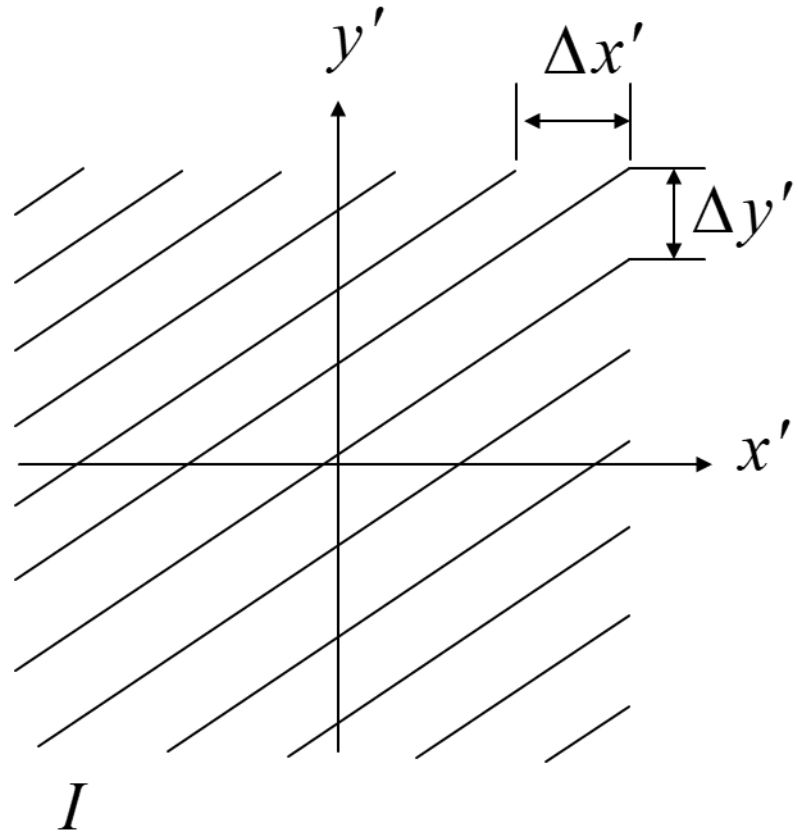
$$\delta(x', y') = kx'(\cos\alpha_1 - \cos\alpha_2) + ky'(\cos\beta_1 - \cos\beta_2) + \varphi_{20} - \varphi_{10}$$

$$I(x', y') = (A_1^2 + A_2^2)\{1 + \gamma\cos[kx'(\cos\alpha_1 - \cos\alpha_2) + ky'(\cos\beta_1 - \cos\beta_2) + \varphi_{20} - \varphi_{10}]\}$$

衬比度: $\gamma = \dfrac{2A_1/A_2}{1 + (A_1/A_2)^2} = \dfrac{2A_1A_2}{A_1^2 + A_2^2}$

$$\delta(x',y') = kx'(\cos\alpha_1 - \cos\alpha_2) + ky'(\cos\beta_1 - \cos\beta_2) + \varphi_{20} - \varphi_{10}$$

$$\delta(x',y') = \begin{cases} 2j\pi & \text{亮纹} \\ (2j+1)\pi & \text{暗纹} \end{cases}$$

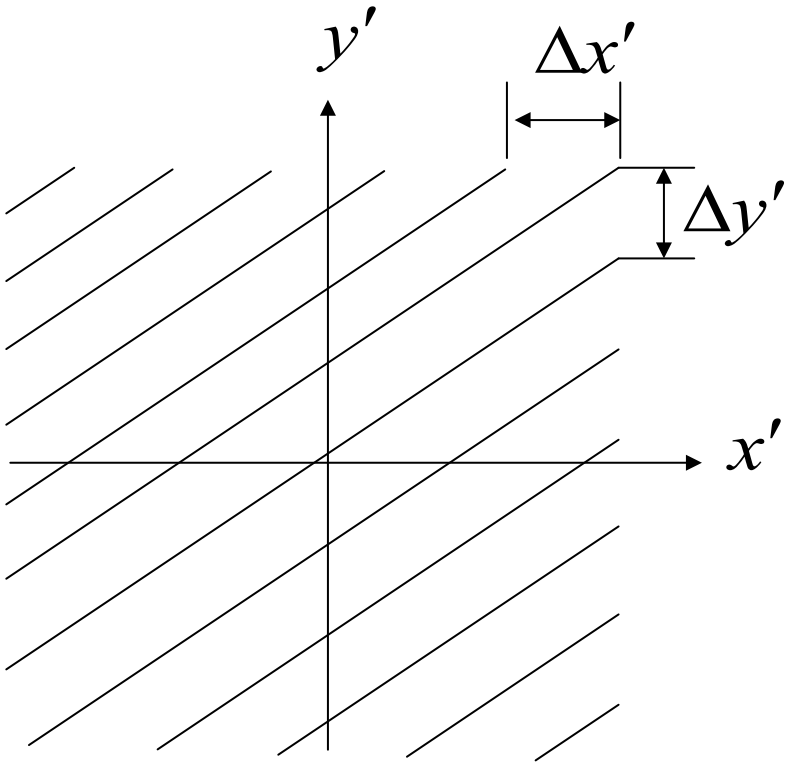


亮暗条纹都是等间隔的平行直线，形成平行直线族，斜率为

$$-\frac{\cos\alpha_2 - \cos\alpha_1}{\cos\beta_2 - \cos\beta_1}$$

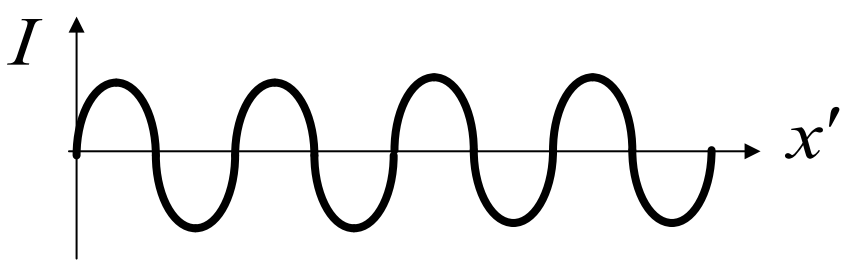
i）干涉条纹的间隔

$$\begin{cases} \Delta x' = \dfrac{2\pi}{k(\cos\alpha_1 - \cos\alpha_2)} = \dfrac{\lambda}{\cos\alpha_1 - \cos\alpha_2} \\ \Delta y' = \dfrac{2\pi}{k(\cos\beta_1 - \cos\beta_2)} = \dfrac{\lambda}{\cos\beta_1 - \cos\beta_2} \end{cases}$$



ii）它们的倒数代表单位长度内的条纹数：**条纹的空间频率**

$$\begin{cases} f_{x'} = \dfrac{1}{\Delta x'} = \dfrac{\cos\alpha_1 - \cos\alpha_2}{\lambda} \\ f_{y'} = \dfrac{1}{\Delta y'} = \dfrac{\cos\beta_1 - \cos\beta_2}{\lambda} \end{cases}$$

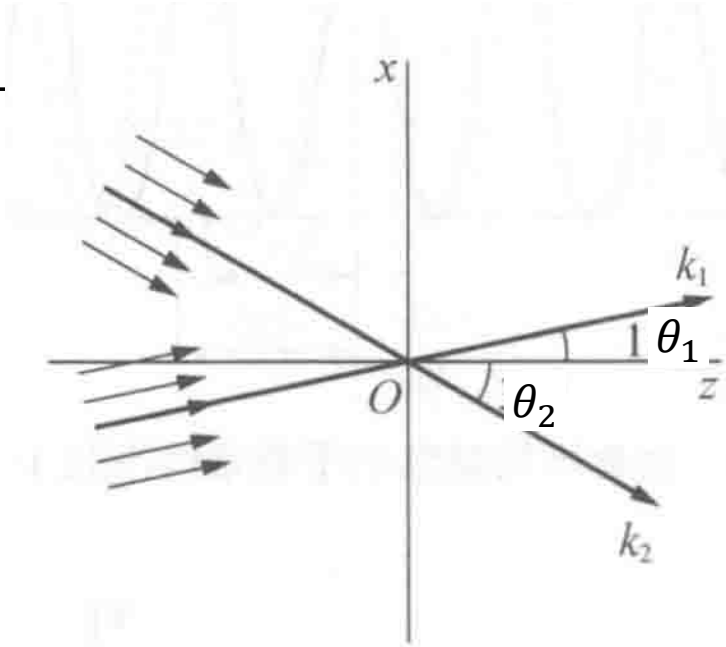


例：两束相干的平行HeNe激光（λ=632.8nm），传播方向在 xz 面内，与 z 轴夹角$\theta_1=\pi/6$，$\theta_2=-\pi/4$，求干涉条纹的间距和空间频率？

间距:

$$\Delta x' = \frac{\lambda}{\cos\alpha_1 - \cos\alpha_2} = \frac{\lambda}{\sin\theta_1 - \sin\theta_2}$$

$$= \frac{632.8nm}{\sin\frac{\pi}{6} + \sin\frac{\pi}{4}} \approx 0.524\mu m$$



空间频率:

$$f_{x'} = \frac{1}{\Delta x'} = \frac{1}{0.524\mu m} \approx 1896mm^{-1}$$

同上题，如想获得低频20mm^{-1} 的干涉条纹，求两平行光的夹角。

空间频率：

$$f_{r'} = \frac{\sin\theta_1 - \sin\theta_2}{\lambda} \approx \frac{\theta_1 - \theta_2}{\lambda} = \frac{\Delta\theta}{\lambda}$$

夹角：

$$\Delta\theta \approx f_{r'}\lambda = 20 \times 632.8 \frac{nm}{mm} \approx 0.013 rad \approx 45'$$

大夹角⇔高空间频率，小夹角⇔低空间频率

作业：p.131： 2, 3, 5, 7,8